

Why We Need a USB Wi-Fi Adapter *(Kali Linux + VirtualBox)*

- Laptop internal Wi-Fi does not support monitor mode inside VirtualBox
- VirtualBox cannot pass internal Wi-Fi to Kali Linux
- Aircrack-ng needs real Wi-Fi signals, not virtual internet
- USB Wi-Fi adapter allows:
-  Monitor Mode,  Packet Capture,  Packet Injection

Important Wi-Fi Terminologies

-  BSSID: Unique MAC address of a Wi-Fi router (router's identity)
-  ESSID: The visible name of the Wi-Fi network
-  CH (Channel): The radio channel on which the Wi-Fi network operates
-  PWR (Power): Signal strength of the Wi-Fi (closer to 0 = stronger)
-  Encryption: Security type used by Wi-Fi (WEP, WPA, WPA2, WPA3)
-  Handshake: Authentication proof exchanged when a device connects to Wi-Fi

 **Which USB Wi-Fi Adapter
to Buy**
(Must-Have Features)

- ✓ Monitor Mode support
- ✓ Packet Injection support
- ✓ Kali Linux compatibility
- ✓ Stable drivers (Atheros / supported Realtek)
- ★ Recommended Adapters
- TP-Link TL-WN722N (v1 ONLY) – best for beginners
- Alfa AWUS036NHA – strong and stable